Know About
Data Science

Who is a Data Scientist?

Data Scientists are analytical experts proficient in solving complex problems by leveraging data insights. Combining IT acumen with business intelligence, they play a pivotal role in driving organisational success. Their expertise spans various industries, including manufacturing, e-commerce, BFSI, healthcare, and transportation.

Curious about the salary range for Data Scientists in India?

The realm of data has undergone a remarkable transformation, presenting abundant career opportunities in recent years. With the technological advancements we're witnessing, companies are eagerly embracing data to derive insights and gain competitive edges. Consequently, there's been a surge in Data Scientist salaries across India, with organisations offering lucrative packages to professionals skilled in data analysis, engineering, and more.

Career Roles	Salary
Data Scientist	15 LPA
Senior Data Scientist	20 LPA
Lead Data Scientist	30 LPA
Data Scientist Manager	50 LPA
Director of Data Scientist	75 LPA
Vice President of Data Scientist	1CR

Why Data Science Matters

Big Data
Leveraging the power of data science enables organisations to harness the potential of big data, turning challenges into opportunities for innovation and strategic decision-making.

Business Intelligence
Data science drives informed decision-making, helping businesses analyse market trends, customer behaviour, and operational efficiency.

Scientific Research
From physics to biology, data science accelerates scientific discovery by unravelling complex datasets and validating hypotheses.

Healthcare
Data science facilitates personalised treatment plans, early disease detection, and targeted therapies, enhancing healthcare outcomes

Policy-making
Governments leverage data science to formulate evidence-based policies and allocate resources effectively, positively impacting communities

Technological Advancements
Data science fuels innovations like artificial intelligence and machine learning, creating intelligent systems with adaptive capabilities.

Skills for Success

- Algorithms, Statistics, Mathematics
- Programming Languages: R, Python, SAS, Hive
- Business Acumen
- Effective Communication

Stay ahead in your career with the requisite skills and expertise demanded by the industry.

Unleash Your Potential! Join the Elite!

Top companies such as **Amazon, Deloitte, EY, IBM, and Microsoft** are actively recruiting Data Scientists, reflecting the high demand and lucrative compensation in this field

Ready to embark on a rewarding career journey in Data Science?

Equip yourself with the skills and knowledge essential for success with Chitti's comprehensive Data Science courses!

Join the Data Science revolution today!

Different Types of Data Science

Descriptive Analytics (Business Intelligence)

- This involves presenting relevant data to the appropriate individuals through dashboards, reports, and emails. Examples include:
- Identifying which customers have stopped using a service.
- Determining which products have been sold in a specific area and whether products of a certain type sell faster.

Predictive Analytics (Machine Learning)

- This involves continuously deploying data science models to make predictions.
- Predicting which customers are likely to stop using a service.
- Estimating the selling price of a product based on its location and other relevant factors.

Prescriptive Analytics (Decision Science)

- This involves using data to assist a company in making decisions. Examples include:
- Determining the best course of action for dealing with customers who are likely to stop using a service.
- Deciding on the most effective marketing strategy for selling a product quickly, based on its location and other relevant factors.

The Standard Data Science Workflow

Data Collection

Compile data from different sources and store it for efficient access

Exploration and Visualization

Explore and visualize data through dashboards

Experimentation and Prediction

The buzziest topic in data science—machine learning!

Your data team members require different skills for different purposes

Data Engineer

This role involves storing and maintaining data using languages such as SQL, Java, Scala, or Python.

Data Analyst

This role involves visualizing and describing data using SQL, Business Intelligence (BI) tools, and spreadsheets

Machine Learning Engineer

This role involves writing production-level code to make predictions using Python, Java, or R.

Data Scientist

This role involves building custom models to drive business decisions using Python, R, or SQL

Python For Data Science - Beginner's Cheat sheet

Why Python ?

Python is recommended for data science due to its simplicity, readability, and ease of learning. It has a rich ecosystem of libraries and frameworks specifically designed for data science, such as NumPy, Pandas, and Scikit-learn. Python's large and active community of developers and data scientists provides ample resources and support. Its flexibility and versatility allow it to integrate with other tools and technologies commonly used in data science. Lastly, Python's popularity and adoption make it a highly sought-after skill in the job market.

Python Cheat Sheet

Accessing Help

- # is used for comments in Python. Everything after # is ignored.
- help(max) displays the documentation for the max function.
- type('a') returns the type of an object, which in this case is str.
  Importing Packages
- Python packages are collections of useful tools developed by the open-source community
- To install a new package, like pandas, use pip install pandas in your command prompt.
- Once installed, import the package

Working Directly

- Python is an interpreted language, meaning you can run code directly in the interpreter or use a script file (.py) to execute code.

Operators

- Arithmetic: +, -, *, /, //, %
- Comparison: ==, !=, >, <, >=, <=
- Logical: and, or, not
- Assignment: =, +=, -=, *=, /=, //=, %=

Getting Started with Lists

- Lists are ordered, mutable, and can contain elements of different data types.
- Example: my_list = [1, 'a', True]

Getting Started with Dictionaries

- Dictionaries are unordered, mutable, and contain key-value pairs.
- Example: my_dict = {'name': 'John', 'age': 30}

Numpy Arrays

- Numpy is a powerful library for numerical computing
- Example: import numpy as np

Math Functions and Methods

- abs(), round(), max(), min(), sum(), len()
- my_list.append(), my_list.insert(), my_list.remove(), my_list.pop()

Getting Started with Characters and Strings

- Strings are immutable sequences of characters.
- Example: my_string = 'Hello, World!'

Getting Started with Data Frames

- DataFrames are 2-dimensional labeled data structures with columns of potentially different types
- Example: import pandas as pd

Example Code

```
# Variables and Data Types
x = 10
y = 'Hello'
z = True

# Lists
my_list = [1, 2, 3, 4, 5]

# Dictionaries
my_dict = {'name': 'John', 'age': 30}

# Numpy Arrays
import numpy as np
my_array = np.array([1, 2, 3, 4, 5])

# Math Functions and Methods
print(abs(-5)) # Output: 5
print(round(3.14159, 2)) # Output: 3.14
print(max(my_list)) # Output: 5
print(min(my_list)) # Output: 1
print(sum(my_list)) # Output: 15
print(len(my_list)) # Output: 5

# Characters and Strings
my_string = 'Hello, World!'
print(my_string[0]) # Output: H
print(my_string[-1]) # Output: !

# Data Frames
import pandas as pd
data = {'Name': ['John', 'Jane', 'Doe'], 'Age': [30, 25, 35]}
df = pd.DataFrame(data)
print(df)
```

Who Can Apply for our

Data Science and Machine Learning Course

- Individuals with a bachelor's degree or final year students keen on learning Data Science.
- IT professionals looking to transition their careers into Data Science.
- Professionals aiming to progress in their IT career.

Program Curriculum

Introduction to Python For Data Science

- Introduction to core Python and Virtual Environment
- NumPy and Pandas Basics
- Python Basics
- Data Preprocessing

Data Visualization with Matplotlib and seaborn

- Introduction to Data Visualization
- Basic Plots with Matplotlib
- Advanced Visualization with Seaborn
- AI VS ML VS DS VS DL

Exploratory Data Analysis (EDA

- Exploratory Data Analysis (EDA)
- Statistical Analysis with SciPy and StatsModels
- Feature Engineering
- Time Series Analysis
- Case Study: EDA on a Real Dataset

Machine Learning Basics

- Introduction to Machine Learning
- Scikit-Learn Basics
- Types of Learning
- Model Evaluation and Cross-Validation
- Introduction to Neural Networks
- Model Deployment Basics
- Case Study: Predictive Modeling

Advanced Topics

- Natural Language Processing (NLP) with NLTK and SpaCy
- Neural Network
- Feature Importance and Model Interpretability
- Time Series Forecasting
- Final Project and Presentation

Things to Master

- Excel
- Python
- Scipy
- Numpy
- Pandas
- Matplotlib
- TensorFlow
- Spark SQL
- Scikit
- Pycharm
- Pylint (Code Coverage)
- Github
- Jenkins

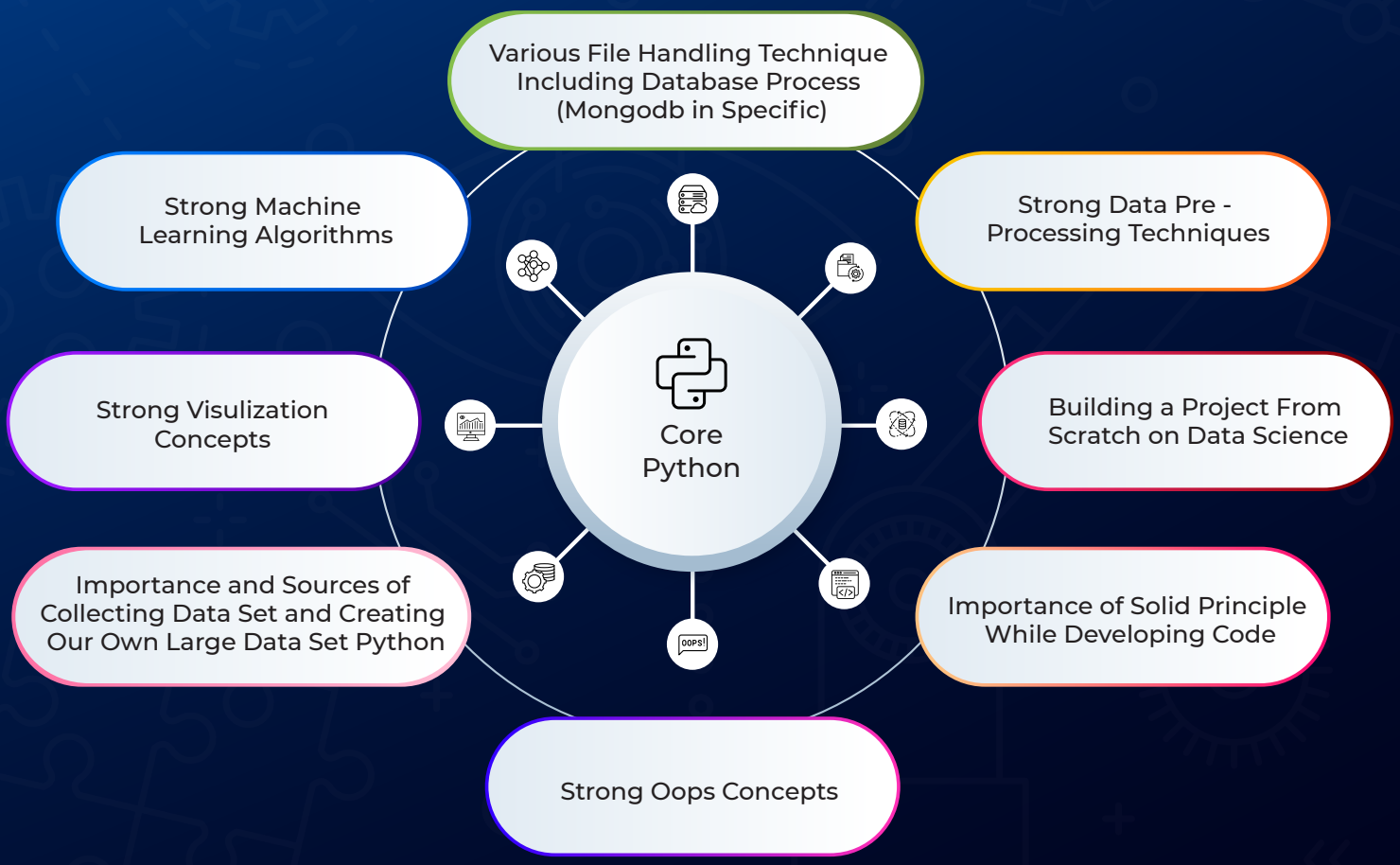
Offerings

- Resume Preparation Module
- Interview Preparation Module
- Linkedin Preparation Module

How to Use

ChatGpt for Data Science

Skills to Master



Project Domain

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